relies on this *probabilistic* variant of validity of a first-order formula φ in a model A. Here, we focus on the *first-order theory of magmas* defined by the signature $\{\diamond : 2\}$ for the binary function $\diamond$. Recall that a magma $(A, \diamond)$ is defined as a set A equipped with a binary operation

$$\diamond \ : \ A \times A \longrightarrow A.$$

A typical first-order predicate φ states the existence of a neutral element:

$$\exists e. \forall x. \quad \big(x \diamond e = x \ \wedge \ e \diamond x = x \big)$$

or the existence of a right residual for any pair of elements x, z of the magma:

$$\forall x. \forall z. \exists y. \quad x \diamond y = z.$$

All equations between magmas can be expressed in that way. Typically, $\mathsf{Eqn}[3721]$ of the equational theory project can be written as the formula

$$\varphi(x, y) \quad \equiv \quad x \diamond y \ = \ (x \diamond y) \diamond (x \diamond y).$$

with two free variables x, y. Now, given a finite magma $(A, \diamond)$ and a first-order formula

$$\varphi(x_1, \ldots, x_n)$$

with n free variables $x_1, \ldots, x_n$, the Stone pairing

$$\langle \, \varphi \, | \, A \, \rangle \ \in \ [0, 1]$$

is the scalar defined as the probability:

$$\frac{\# \left\{ \, (a_1, \ldots, a_n) \in A^n \mid \varphi(a_1, \ldots, a_n) \, \text{holds} \, \right\}}{\# A^n}$$

that a n-tuple of elements

$$(a_1, \ldots, a_n) \in A^n$$

satisfies the n-ary formula φ. Here we write $\# A$ for the cardinal of a finite set A. By way of illustration, observe that

$$A \ \vDash \ \forall \mathbf{x}, \varphi(\mathbf{x}) \quad \Longleftrightarrow \quad \langle \ \varphi(\mathbf{x}) \ | \ A \ \rangle \ = \ 1$$

where we write $A \vDash \psi$ when a first-order formula ψ is satisfied by a given magma A, and use the notation $\mathbf{x} = x_1, \ldots, x_n$ for the sequence of free variables of the formula φ. From this follows in particular that

$$\langle \ x = x \ | \ A \ \rangle \ = \ 1$$

for every finite magma A. Fractional values different from 0 and 1 are the most common in practice. For instance, note that

$$\langle \ x = y \ | \ A \ \rangle \ = \ \frac{1}{\# A}$$

for every finite magma A, since there is a probability of one divided by the cardinal of A that two elements $a_1, a_2 \in A$ happen to be equal.

5 The feature space of equational theories

In this section, we explain how we use the probabilistic concept of Stone pairing in order to construct our latent space of equational theories.

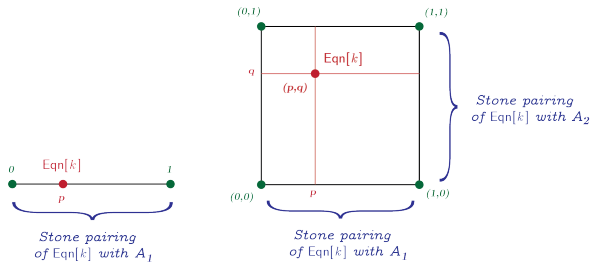

Figure 1: How each equation $\mathsf{Eqn}[k]$ is assigned a position in the hypercube $\mathbb{H}_n$ of dimension $n = \#\mathsf{Mags}$ the number of sampled finite magmas $A_1, \ldots, A_n$.

5.1 The generation process

We proceed in the following way. We start by generating a large number $n = \#\mathsf{Mags}$ of finite magmas $A_1, \ldots, A_n$ of a fixed size N. For simplicity, we define them with the same underlying set $\{0, \ldots, N-1\}$. Each magma A_ℓ for $1 \leq \ell \leq n$ is thus generated as a two-dimensional array of size $N \times N$, where each entry is a randomly chosen value between 0 and $N - 1$. This two-dimensional array describes the multiplication table of the magma $A_\ell = \{0, \ldots, N-1\}$ and thus completely characterizes it. We typically work with a sample of $n = 1000$ magmas of size N between 4 and 16. These numbers $n = 1000$ and $4 \leq N \leq 16$ appear reasonable from a computational point of view and sufficient for our purposes.

5.2 The feature hypercube $\mathbb{H}_n = [0, 1]^n$

We define a matrix of size $\#\mathsf{Eqns} \times \#\mathsf{Mags}$

$$R \quad = \quad \big(p_{k,\ell} \big)_{k \in \mathsf{Eqns}, \ell \in \mathsf{Mags}} \tag{1}$$

where $\#\mathsf{Eqns} = 4964$ is the number of equations and $n = \#\mathsf{Mags} = 1000$ is the number of sampled magmas of size N. Each entry $p_{k,\ell}$ of the matrix R is defined as the probability

$$p_{k,\ell} \quad = \quad \langle \, \mathsf{Eqn}[k] \, | \, A_\ell \, \rangle$$

given by the Stone pairing of $\mathsf{Eqn}[k]$ with the finite magma A_ℓ. Every equation $\mathsf{Eqn}[k]$ induces a line or vector of n probabilities

$$\big(p_{k,1} \ , \ \cdots \ , \ p_{k,n} \big) \ \in \ [0, 1]^n$$

which defines an element of the hypercube $\mathbb{H}_n = [0, 1]^n$ of dimension $n = \#\mathsf{Mags}$, see Fig. 1 for an illustration.

5.3 Stone spectrum

The hypercube $\mathbb{H}_n = [0, 1]^n$ comes equipped with an action